

Evaluating the Impact of Florida's 2010 Opioid Control Policies

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Target Stakeholder

Our target stakeholder is a **State Representative's Policy Advisor** who evaluates legislative options to address opioid-related harms. Their primary question is whether supply-side prescription restrictions—like Florida's 2010 policy—can meaningfully reduce opioid-related mortality. They are also concerned that limiting legal access may push dependent individuals toward more dangerous illicit opioids such as heroin or fentanyl. The advisor has a solid policy background and can interpret high-level empirical findings to inform legislative recommendations.

Executive Summary

In the United States, many states continue to face rising opioid-related harms, much of which stem from the rapid growth in prescription opioid use. Policymakers are increasingly looking for approaches that can reduce misuse and overdose risk while still ensuring appropriate access for patients who rely on these medications.

To contribute to this conversation, we examine Florida's 2010 prescription-control regulations, one of the earliest statewide efforts to address unusually high levels of opioid dispensing. Our analysis compares Florida's trends in opioid shipments and overdose mortality with those in similar states (Pennsylvania, Illinois, and Oklahoma) that did not adopt such regulations. This comparison helps distinguish the policy's effect from broader national trends that might have influenced all states equally.

We analyze county-level data on opioid shipments from the DEA's ARCOS system and overdose deaths from US Vital Statistics records, covering 2006-2015. Opioid shipments are measured in morphine milligram equivalents (MME), a standardized unit that allows comparison across different opioid types based on their relative potency.

The results indicate that the policy was associated with meaningful short-term reductions in both opioid shipments and overdose mortality. Shipments fell substantially after the intervention, and mortality declined in the early post-policy years.

However, long-term effects are more difficult to assess. The emergence of illicit fentanyl after 2013 created a new and far more potent driver of overdose deaths, complicating evaluations of sustained policy impact. For this reason, states considering similar measures should view supply-side regulations as one component of a broader strategy, rather than a standalone solution.

Background

Over the past two decades, many U.S. states have experienced steady increases in opioid-related harm, much of it linked to rising prescription volumes. National data show that between 1999 and 2010, opioid prescribing rates quadrupled, and overdose deaths from prescription opioids rose alongside them.¹ These trends have prompted legislators in several states to re-examine whether limiting excessive prescribing could help reverse mortality patterns.

Florida's experience offers a useful point of comparison. Prior to 2010, the state reported some of the highest levels of opioid dispensing in the country; 98 of the top 100 oxycodone-dispensing physicians

¹ Centers for Disease Control and Prevention. (2011). *Vital Signs: Overdoses of Prescription Opioid Pain Relievers—United States, 1999–2008*. MMWR Morbidity and Mortality Weekly Report, 60(43), 1487–1492.

nationwide were practicing in Florida at the time.² In response, the legislature introduced a set of supply-side prescription regulations aimed at reducing unusually high levels of opioid availability.

For states considering similar measures today, a central concern is whether reducing legal prescriptions actually lowers overdose deaths. One complication is that people already dependent on opioids may turn to illicit drugs if prescriptions become harder to obtain. National surveillance data indicate that illicit fentanyl began driving a new rise in overdose deaths around 2013, largely because of its extreme potency; even very small doses can be fatal, with some estimates placing fentanyl at 50 to 100 times stronger than commonly prescribed opioids.³⁴

Within this broader context, Florida's 2010 regulations offer a relevant example for evaluating whether supply-side prescription controls can meaningfully reduce opioid shipments and improve mortality—questions of direct interest to state-level legislative advisors considering similar policy interventions.

Did Florida's 2010 Opioid Policy make a difference?

A simple pre-post analysis is a natural starting point for evaluating whether Florida's 2010 reform coincides with changes in opioid shipments or overdose mortality. Comparing outcomes before and after the policy can highlight broad shifts in the direction we might expect. However, this type of analysis is insufficient for our purposes because it cannot separate the effect of the reform from other forces changing at the same time. National prescribing guidelines, changes in pharmaceutical marketing, or economic conditions could all influence shipment levels regardless of Florida's policy. A purely within-state comparison would also risk confounding the reform's effect with Florida-specific trends that were already emerging before the law took effect.

To address these challenges, we turn to a difference-in-differences approach, which compares how outcomes change over time in Florida relative to how they change in similar states without the policy. By examining differences in trends across states, this method helps isolate the impact of the reform from nationwide shocks or persistent differences across regions.

A difference-in-differences design strengthens the analysis by comparing Florida to similar states that did not adopt the reform. Observing the same period in Pennsylvania, Illinois, and Oklahoma allows us to account for national shocks that influence all states and to hold constant time-invariant differences across counties, such as health infrastructure, demographic baselines, or local regulatory environments. Although no method removes every possible confounder, the DiD framework provides a more credible counterfactual than descriptive statistics. The full DiD regression equation is presented in Appendix B.

To ensure this comparison is valid, we first examine whether Florida and the comparison states followed similar trends before 2010. If the states were already converging or diverging, the post-policy estimates might simply capture those pre-existing patterns. For analytical clarity, we also limit our sample to counties with an average population of at least 350,000 to ensure more stable outcome measures. Our diagnostics, shown in Appendix B, indicate that mortality trends track closely among the states, while shipment patterns display some pre-policy convergence that informs how we interpret the results.

² Johnson, H., Paulozzi, L., Porucznik, C., Mack, K., & Herndon, J. (2014). *Decline in drug overdose deaths after state policy changes—Florida, 2010–2012*. MMWR Morbidity and Mortality Weekly Report, 63(26), 569–574.

³ Texas Health and Human Services. (2022). *Fentanyl: One Pill Kills*. Office of Mental Health and Substance Use. <https://www.hhs.texas.gov/services/mental-health-substance-use/mental-health-substance-use-resources/fentanyl-one-pill-kills>

⁴Centers for Disease Control and Prevention. (2022). *Understanding the Opioid Overdose Epidemic*. National Center for Injury Prevention and Control. <https://www.cdc.gov/overdose-prevention/about/understanding-the-opioid-overdose-epidemic.html>

Opioid shipments

The trend plot of opioid shipments (MME per 1,000 residents) shows a clear decline in Florida beginning in 2010.



Figure 3: Opioid Shipments - per 1000 Residents

This pattern is consistent with the results of our regression analysis.

Outcome	Intercept	DiD (Post × Treated)	Std. Error	p-value	Controls	N	R ²
MME per 1,000 population	1211.547000	-155.105000	80.435000	0.054000	Yes	320	0.871000

Controls include county and year fixed effects. Standard errors clustered by county.

Table 1: Difference-in-Differences Regression: Opioid Shipments

In the shipment regression, the DiD coefficient on the treatment–post interaction is negative and marginally significant ($p = 0.054$). While this does not reach conventional significance at the 5% level, the direction and magnitude of the estimate are consistent with a policy-induced reduction in shipments. This implies that, after 2010, opioid shipments in Florida’s large counties fell by more than in the matched control counties, even after accounting for county fixed effects and common year shocks. The magnitude of the effect is meaningful: the estimated policy impact corresponds to a sizable reduction in MME per 1,000 residents relative to the pre-policy baseline, consistent with the sharp post-2010 decline seen in the descriptive trends.

Overdose Mortality

The mortality rates in Florida and the control states over the past ten years are shown below.

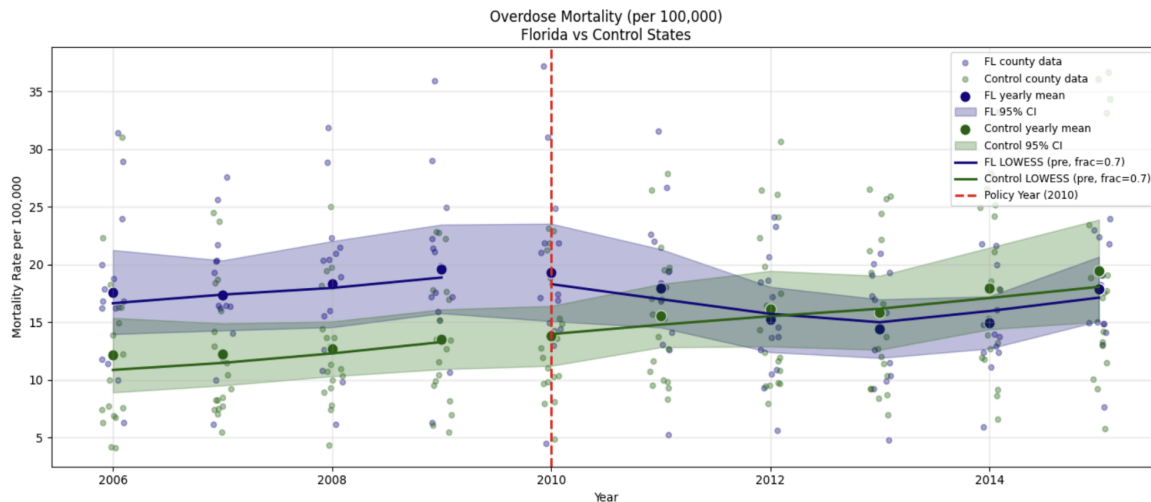


Figure 4: Opioid Related Deaths - per 100000 Residents

For overdose mortality, the DiD coefficient is also negative and statistically significant.

Outcome	Intercept	DiD (Post $\times$ Treated)	Std. Error	p-value	Controls	N	R ²
Overdose deaths per 100,000	22.523000	-5.399000	1.046000	0.000000	Yes	320	0.826000

Controls include county and year fixed effects. Standard errors clustered by county.

Table 2: Difference-in-Differences Regression: Overdose Mortality

This indicates that, following the policy, drug-related deaths declined more in Florida than in the control states, conditional on county and year fixed effects. Taken at face value, the estimates suggest a reduction of several deaths per 100,000 population attributable to the policy, aligning with the drop in mortality observed in the early post-policy years.

These DiD results are broadly consistent with the pre-post analysis: Florida's 2010 opioid regulations are associated with lower opioid shipments and reductions in overdose mortality relative to similar control counties. However, the event study evidence indicates that pre-policy trends in shipments were not perfectly parallel, with Florida "catching up" to control states even before the 2010 intervention. For mortality, the pre-policy differences are smaller but still not perfectly flat. As a result, while the DiD estimates provide reasonably strong evidence that the policy reduced shipments and likely contributed to lower mortality, they should be interpreted with caution rather than as definitive causal estimates.

Robustness Check

To ensure the findings are not unique to Florida, we broaden the analysis to include Georgia and Maryland, which introduced similar opioid regulations soon after Florida did. We compare counties in these three policy states to large counties in Pennsylvania, Illinois, and Oklahoma, which did not adopt comparable supply-side opioid legislation during this period. We continue to focus on counties with at least 350,000 residents to keep the comparison consistent and to avoid results being driven by very small counties.

Shipments

The opioid shipments in treatment states and the control states over the past ten years are shown below.

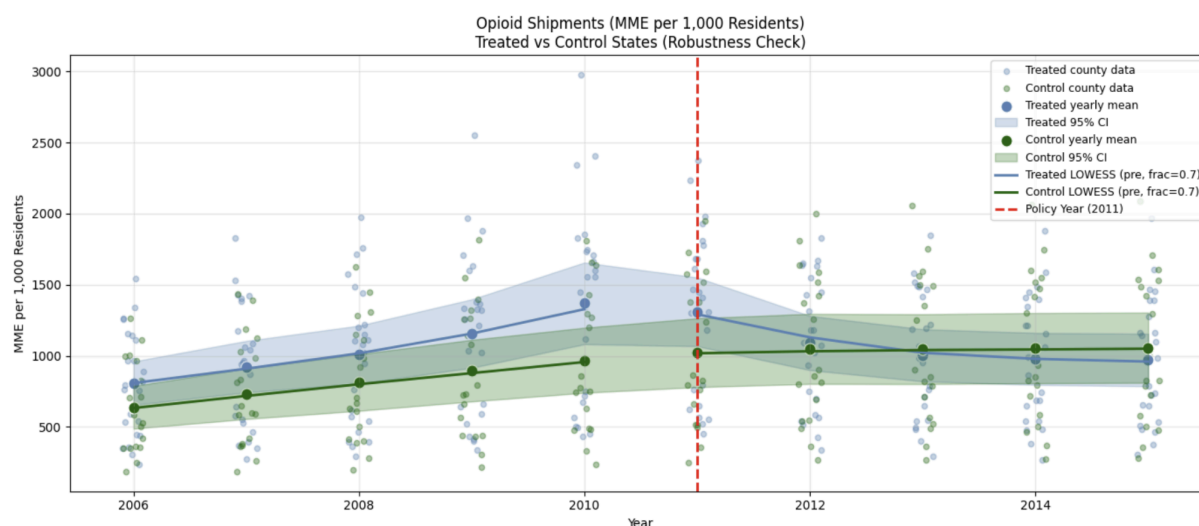


Figure 5: Opioid Shipments (Treated vs Control) - per 100000 Residents

In this expanded specification, the DiD interaction for opioid shipments is negative (approximately -117 MME per 1,000 residents) and marginally statistically significant at the 10 percent level.

Outcome	Intercept	DiD (Post $\times$ Treated)	Std. Error	p-value	Controls	N	R ²
MME per 1,000 population	1204.349000	-116.867000	64.691000	0.071000	Yes	410	0.881000

Controls include county and year fixed effects. Standard errors clustered by county.

Table 3: Difference-in-Differences Regression: Opioid Shipments (Robustness check)

The expanded comparison shows the same core pattern as the main analysis. After the regulations went into effect, opioid shipments dropped more in the policy states than in the comparison group. The estimated reduction is roughly 100 to 120 MME per 1,000 residents. While the statistical strength is slightly weaker than in the Florida-only model, the direction and size of the effect remain consistent. Taken together, this supports the conclusion that supply-side regulations were followed by meaningful reductions in the volume of prescription opioids across multiple states.

Overdose mortality

The mortality rates in treatment states and the control states over the past ten years are shown below.

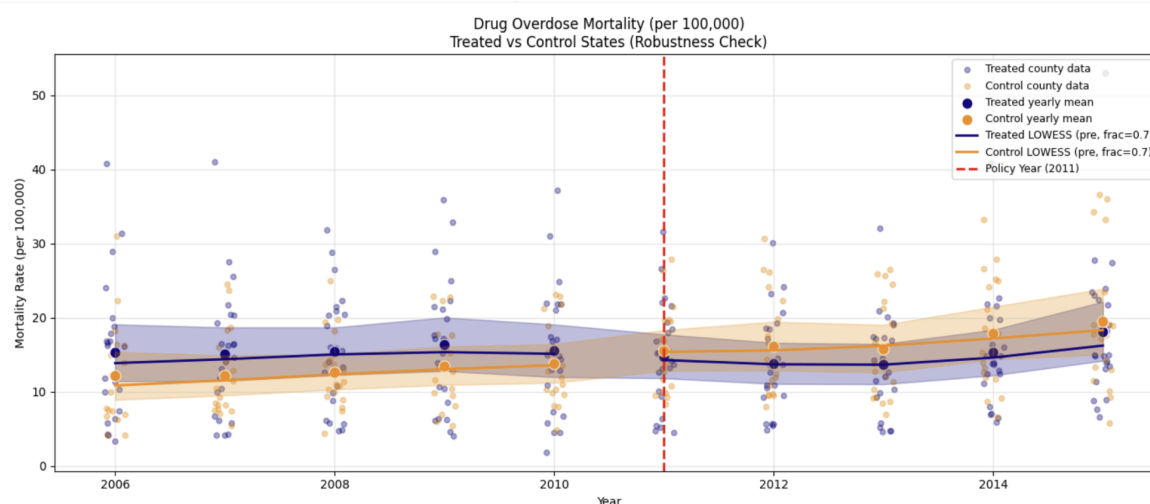


Figure 6: Opioid related deaths (Treated vs Control) - per 100000

For overdose mortality, the robustness DiD coefficient is around -4.15 and highly statistically significant ($p < 0.001$).

Outcome	Intercept	DiD (Post $\times$ Treated)	Std. Error	p-value	Controls	N	R ²
Overdose deaths per 100,000	21.935000	-4.154000	0.991000	0.000000	Yes	410	0.842000

Controls include county and year fixed effects. Standard errors clustered by county.

Table 4: Difference-in-Differences Regression: Overdose Mortality (Robustness check)

A similar pattern appears for overdose deaths. Counties in Florida, Georgia, and Maryland experienced larger declines in opioid-related mortality than counties in the comparison states. On average, deaths fell by a little over four fatalities per 100,000 residents in the policy states relative to the controls. This reinforces the idea that early reductions in prescription opioid availability were accompanied by short-term declines in overdose mortality.

Limitations

While the results show clear declines in prescription opioid shipments after the policies were adopted, interpreting the changes in overdose deaths requires more caution. Mortality does fall in the first few years after the policy, but the trend shifts sharply beginning around 2013. This turning point reflects a major national change in the drug supply that was unrelated to state regulations.

Beginning in 2013, the United States saw a rapid surge in deaths linked to illicitly manufactured fentanyl. This substance is cheaper, far more potent, and considerably more dangerous than prescription opioids. Its spread affected nearly every state regardless of policy. As fentanyl became more common, overdose deaths rose nationwide, including in states that had reduced prescription opioid access.

Because this national fentanyl wave occurred at the same time that mortality in Florida and other policy states began to rise again, it becomes difficult to separate the long-term effects of the policy from the influence of this broader shift in the drug market. The increase in deaths after 2013 reflects forces that the state-level regulations could not realistically control.

In short, the evidence strongly supports the conclusion that the policies reduced prescription opioid shipments and contributed to early mortality declines. However, the long-term mortality trend should be understood as the combined result of the state regulations and the nationwide emergence of fentanyl, rather than as a clean estimate of the policy's effect on its own.

Appendix A: Pre-Post Analysis in Florida

This appendix presents a pre-post comparison of opioid shipments and overdose mortality in Florida. While the main analysis compares Florida to control states using difference-in-differences, this section examines Florida's trends in isolation to illustrate the raw patterns in the data.

Methodology

We retain data at the county-year level rather than aggregating to state-year totals. This approach preserves the cross-county variance, which is essential for accurate standard error estimation. We fit separate LOWESS (Locally Weighted Scatterplot Smoothing) curves for the pre-policy period (2006-2009) and the post-policy period (2010-2015). LOWESS fits a smooth curve by performing weighted local polynomial regressions at each data point, where observations closer to the target point receive higher weights. The smoothing parameter ($\text{frac} = 0.7$) specifies the fraction of data used in each local regression. Fitting separate curves prevents the smoother from treating the policy-induced discontinuity as noise to be smoothed over. The shaded bands represent 95% confidence intervals for the annual mean, calculated using the standard error of the mean across counties.

Opioid Shipments (MME per 1000 Population)

Figure A1 shows opioid shipments measured in Morphine Milligram Equivalents (MME) per 1000 residents. Gray points show county-level observations; blue points indicate annual means; blue curve shows separate LOWESS fits for pre- and post-policy periods; shaded area represents 95% CI for the mean.

Pre-policy period (2006-2009): Opioid shipments rose steadily from approximately 1000 MME per 1000 in 2006 to 1500 in 2009, representing a 50% increase over this period.

Policy year (2010): Shipments peaked at approximately 1750 MME per 1000, coinciding with Florida's regulatory changes. This spike may reflect stockpiling behavior in anticipation of stricter enforcement.

Post-policy period (2010-2015): Following the policy, shipments declined sharply to approximately 1000 MME per 1000 by 2015, namely a 43% reduction from the 2010 peak and return to near-2006-level.

The visual discontinuity between the pre- and post-policy LOWESS curves clearly illustrates the structural break in opioid supply trends associated with Florida's intervention.

Overdose Mortality (Deaths per 100000 Population)

Figure A2 shows drug overdose mortality rates per 100,000 population. The pattern differs notably from opioid shipments.

Pre-policy period (2006-2009): Mortality rates were relatively stable, fluctuating between 15 and 20 deaths per 100000 with a slight upward trend.

Initial post-policy decline (2010-2013): Following the policy, mortality declined to approximately 14-17 deaths per 100000 by 2013, suggesting a beneficial effect of reduced opioid supply.

Subsequent rebound (2014-2015): Mortality rose again to approximately 18 deaths per 100000 by 2015. This rebound coincides with the national rise in illicit fentanyl-related deaths beginning around 2013, suggesting that some users may have substituted toward more dangerous illicit opioids as prescription access became restricted.

The wider confidence intervals and greater scatter of county-level points indicate substantial heterogeneity in mortality outcomes across Florida counties.

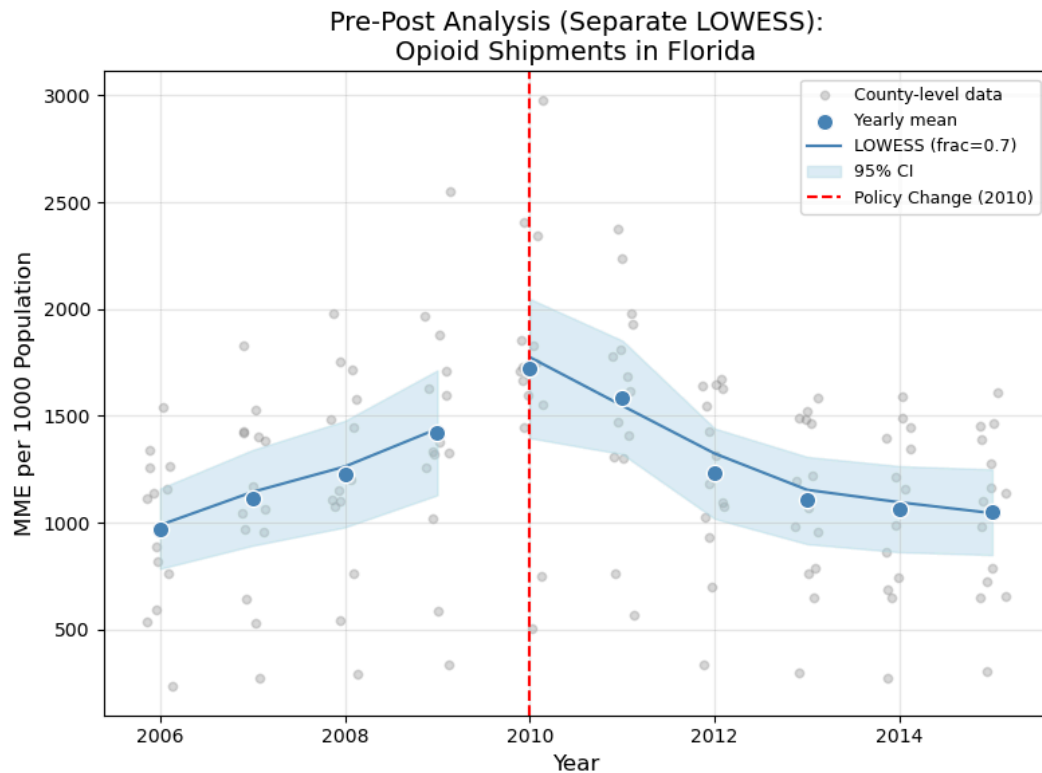


Figure A1: MME per 1000 Population in Florida, 2006-2015 (LOWESS)

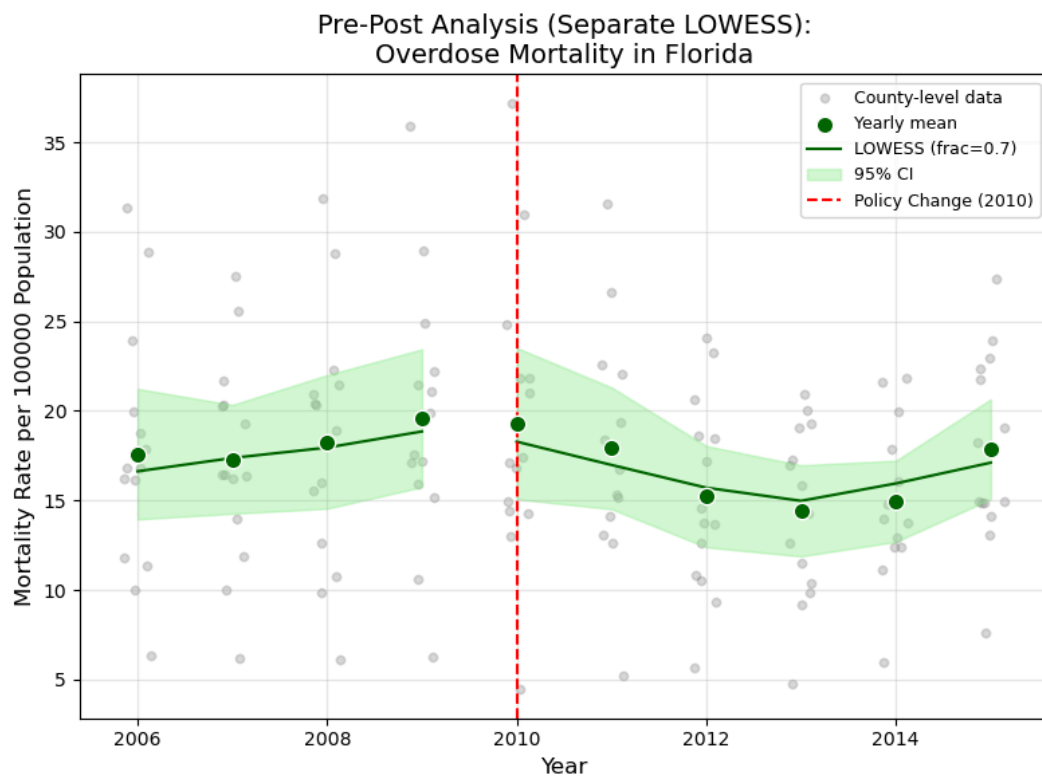


Figure A2: Overdose Mortality Rate per 100000 Trend in Florida, 2006-2015 (LOWESS)

Interpretation and Limitations

The pre-post analysis reveals a clear structural break in opioid shipments coinciding with Florida's 2010

policy intervention. The divergent patterns between shipments and mortality highlight an important policy implication: while supply-side restrictions can effectively reduce prescription opioid distribution, mortality outcomes may be influenced by substitution effects as users transition to illicit alternatives.

Several limitations should be noted. First, pre-post analysis cannot establish causality, as other contemporaneous factors may have contributed to observed trends. Second, the analysis is restricted to counties with average population of at least 350000 to reduce noise in mortality rate estimates. Results may not generalize to smaller or rural counties, which could have different opioid use patterns. Third, the mortality rebound after 2013 suggests that the longer-term effects of supply restrictions may differ from short-term impacts.

For causal identification, the main text presents difference-in-differences estimates comparing Florida to appropriate control states.

Appendix B: Technical Validation of the Analysis Design

The difference-in-differences approach relies on a critical assumption: before the policy, Florida and the control states (Pennsylvania, Illinois, and Oklahoma) were following similar trends. If they were not, the post-policy divergence might reflect pre-existing differences rather than the policy's effect. To test this, we use an event study analysis comparing Florida to control states in each year separately. The key question is: were there meaningful differences between Florida and control states before 2010?

Sample and Variables

The sample includes Florida (treated) and three control states: Pennsylvania, Illinois, and Oklahoma. We created an event time variable by subtracting 2010 from the calendar year, so event_time = -1 corresponds to 2009 (the reference year), and event_time = 0 corresponds to 2010 (the policy year).

Regression Specification

We estimated the following event study model:

$$Y_{it} = \theta_i + \sum_{k \neq -1} \alpha_k 1\{event_time = k\} + \beta Treated_i + \sum_{k \neq -1} \gamma_k (1\{event_time = k\} \times Treated_i) + \varepsilon_{it}$$

where Y_{it} is the outcome in county i and year t ; event_time = k indexes years relative to the policy year, with $k = -1$ as the baseline; α_k are event-time fixed effects absorbing shocks common to all counties in each relative year; β captures any average level difference between Florida and the control states; γ_k captures the event-time-specific treatment effect, i.e. the difference between (Florida - Controls) in event year k and (Florida - Controls) in $k = -1$; θ_i are county fixed effects controlling for all time-invariant county characteristics.

We do not include separate calendar-year fixed effects because event-time indicators are collinear with calendar years once the time axis is re-centered around the policy year. In this specification, C(event_time_cat) fully absorbs common time shocks in Python

Results: Opioid Shipments

Figure B1 presents the event study results for opioid shipments (MME per 1000 population). Each point shows the estimated difference between Florida and control states in that year, relative to the baseline year (2009). The pre-policy coefficients (event time -4 to -2) are negative and statistically different from zero, with confidence intervals that do not include zero. This pattern indicates that, relative to the base pre-policy year (2009), Florida's opioid shipments were catching up to (and eventually exceeding) those of the control states even before the policy intervention. This challenges the parallel-trends assumption.

At event time 0, we see a sharp rise, followed by a steep and persistent decline in the estimated effects. By 2015 (event time +5), Florida's MME is roughly 200-250 units per 1000 population lower than control states, relative to the base year. This pattern indicates a meaningful long-run reduction in shipments after the policy. However, because the pre-policy trajectory is not flat, we cannot fully rule out the possibility that part of the post-policy decline reflects a correction of those earlier dynamics rather than a purely causal policy effect. These results should therefore be interpreted with caution.

Results: Overdose Mortality

Figure B2 shows the event study estimates for drug-related mortality rates. The pre-policy coefficients (event time -4 to -2) are positive, clustered around 2 deaths per 100000. The earliest year (2006) has a confidence interval that narrowly excludes zero, while the remaining pre-policy years have confidence intervals that marginally include zero. This pattern suggests that Florida had a somewhat higher mortality level than control states before the policy, though the differences are modest and only borderline significant.

After the policy, the estimated effects gradually decline: the coefficients turn negative by event time +2 and reach about -2 to -2.5 deaths per 100000 by event time +4. This downward trajectory indicates a positive policy impact on mortality. However, because the pre-treatment trend is not perfectly flat, we cannot definitively attribute the entire post-policy decline to the policy itself.

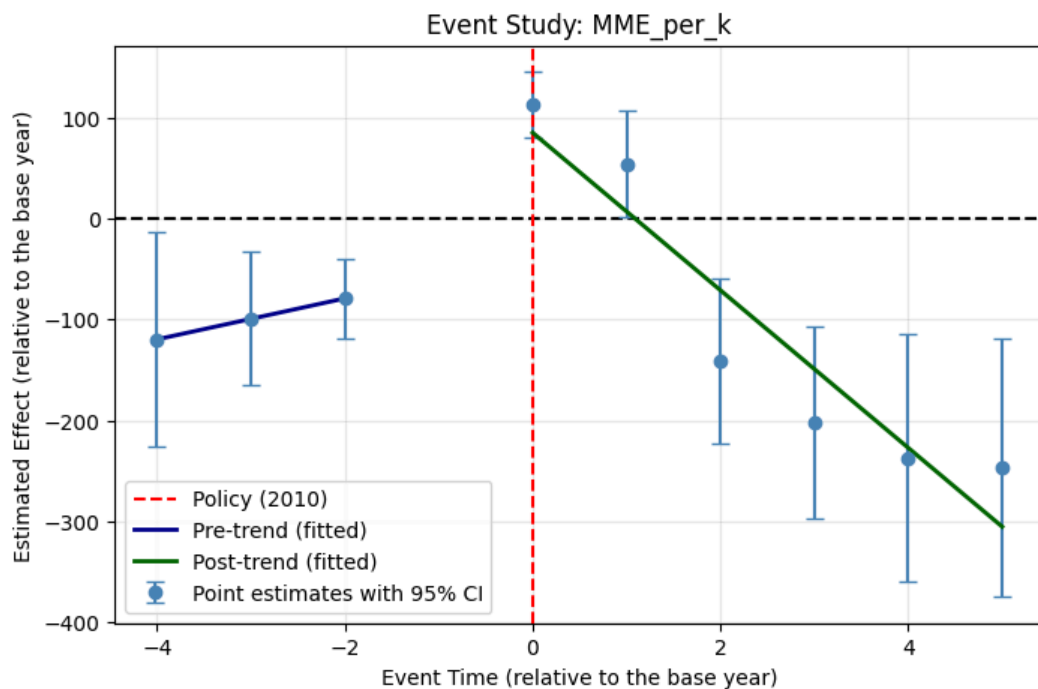


Figure B1: Event Study - MME per 1000 Population

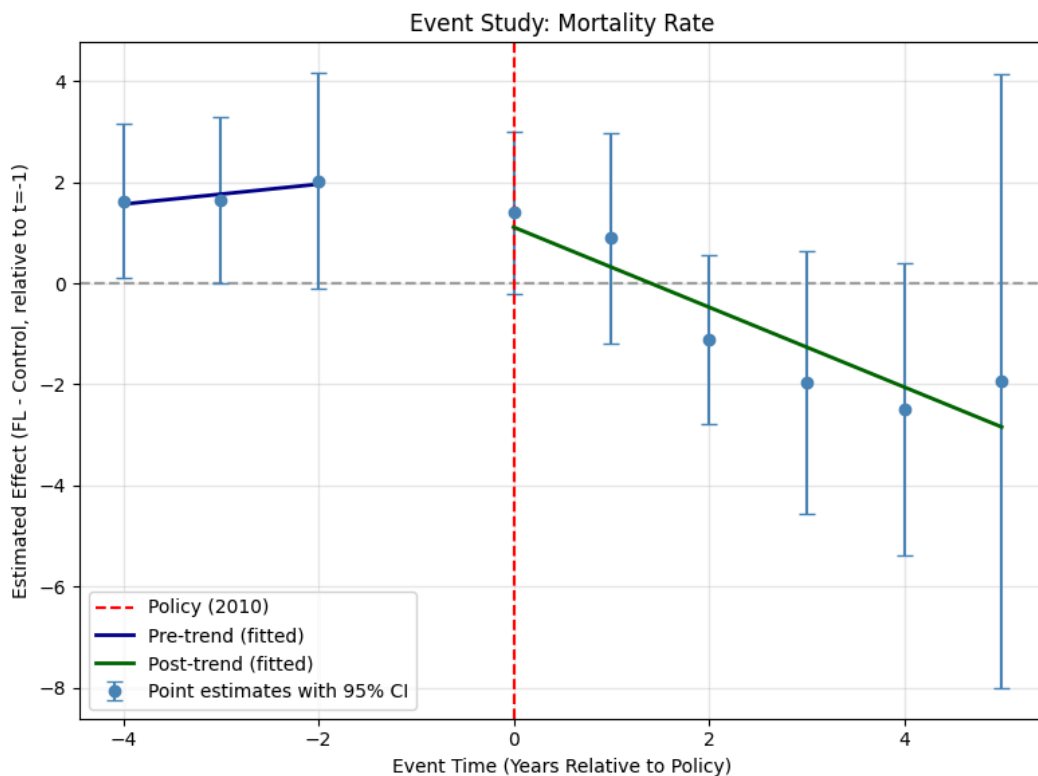


Figure B2: Event Study - Mortality Rate per 100000 Population

Interpretation

Under the parallel trends assumption, pre-policy coefficients ($\gamma < 0$) should be close to zero. For mortality, this assumption is roughly supported: pre-policy coefficients are near zero with confidence intervals that include zero. For MME, the pre-policy pattern shows more substantial divergence. These findings inform the interpretation of the main difference-in-differences results: the shipment estimates should be viewed with some caution due to pre-trend concerns, while the mortality estimates rest on somewhat stronger ground.

Difference-in-Differences Model Specification

For each outcome Y_{it} (opioid shipments or overdose mortality) in county i and year t , we estimate the following difference-in-differences specification:

$$Y_{it} = \alpha_i + \lambda_t + \beta Treated_i \times Post_t + E_{it}$$

Where α_i are county fixed effects λ_t are year fixed effects, $Treated_i$ indicates Florida counties, $Post_t$ indicates post-2010 years, and β is the DiD estimate of the policy's effect. Standard errors are clustered at the county level.

Appendix C: Data Sources and Processing

Data Sources

This study draws on multiple data sources to construct a county-level panel dataset covering the period from 2006 to 2015. The primary data on prescription opioid shipments was obtained from the Washington Post, which acquired the data through a Freedom of Information Act (FOIA) request to the US Drug Enforcement Administration's Automation of Reports and Consolidated Orders System (ARCOS). This dataset includes the calculated base weight in grams and the morphine milligram equivalent (MME) conversion factor for each opioid type.

Mortality data was obtained from US Vital Statistics records, specifically the Underlying Cause of Death files for each year from 2006 to 2015. These files contain county-level death counts classified by drug and alcohol-induced causes. The analysis focuses on deaths classified under codes D1 (unintentional drug poisoning), D2 (suicide by drug poisoning), D4 (drug poisoning of undetermined intent), and D9 (other drug-induced causes), which together capture the spectrum of drug overdose deaths.

County-level population data was obtained from the US Census Bureau. For the years 2006 to 2010, population estimates were drawn from the County Intercensal Tables: 2000-2010. For the years 2010 to 2015, data came from the County Intercensal Population Totals: 2010-2020. These intercensal estimates provide annual population figures for each county, which are necessary for calculating per capita measures.

Socioeconomic and demographic characteristics were obtained from the 2005-2009 American Community Survey (ACS) 5-Year Estimates via the Census Bureau's API. The specific variables extracted include the poverty universe total, the count of individuals below the poverty level, the total population for the race universe, the non-Hispanic white population, the non-Hispanic Black population, and the Hispanic or Latino population of any race.

Sample Construction

The analysis focuses on Florida as the treatment state and a pool of fourteen potential control states: Georgia, Iowa, Illinois, Indiana, Kansas, Maryland, Michigan, Missouri, North Carolina, New Hampshire, Oklahoma, Pennsylvania, Utah, and Wisconsin. These states were selected as candidates for the control group based on demographic similarity and the absence of major opioid policy interventions during the study period.

To ensure sufficient statistical power and reliable estimation of county-level trends, the analysis restricts the sample to counties with populations exceeding 350,000 residents. This population threshold results in a final sample of 62 counties across 15 states, comprising 14 Florida counties (the treatment group) and 48 counties from the potential control states (the control pool).

Morphine Milligram Equivalent (MME) per 1,000 Population

The volume of opioid shipments was measured using morphine milligram equivalents (MME), a standardized metric that allows comparison across different opioid types based on their relative potency. For each county-drug-year observation in the ARCOS data, the MME was calculated by multiplying the base weight in grams by the corresponding MME conversion factor:

$$\text{MME} = \text{calc_base_wt_in_gm} \times \text{mme_conversion_factor}$$

The county-level MME for each year was then computed by aggregating (summing) the MME values across all opioid types within each county-year combination. To account for differences in county population sizes, this total was normalized to create a per capita measure:

$$\text{MME per 1000 population} = (\text{Total MME} / \text{County Population}) \times 1000$$

This measure represents the total morphine-equivalent milligrams of opioids shipped to each county per

1000 residents, providing a standardized indicator of opioid supply intensity.

Drug Overdose Mortality Rate per 100000 Population

The mortality outcome was constructed from the CDC WONDER Underlying Cause of Death files. For each year, death records were filtered to retain only those classified under drug-induced cause codes D1, D2, D4, and D9. County names were extracted and cleaned by removing state suffixes and the word “County,” and then standardized to uppercase format with spaces removed to facilitate merging with other datasets.

Annual county-level death counts were computed by summing deaths across all four drug-induced cause categories. For county-years with no reported deaths (either due to suppression or true zeros), the death count was set to zero. The mortality rate was then calculated as:

$$\text{Mortality Rate} = (\text{Drug Deaths} / \text{County Population}) \times 100000$$

This measure represents the number of drug overdose deaths per 100000 county residents.

Appendix D: Control State Selection

Eligible Control Pool

We excluded states that implemented mandatory Prescription Drug Monitoring Programs (PDMPs) or Pain Clinic Laws before 2015, following Dowell et al. and Wen et al.⁵ We also excluded small or sparsely populated states due to data suppression issues. The remaining states formed the eligible control pool.

Matching Variables

The propensity score model included the following pre-treatment (2006-2009) covariates:

- Poverty rate (from ACS 2005-2009)
- Racial composition (percent non-Hispanic white, percent Hispanic)
- Mean MME per 1000 population
- Mean overdose mortality rate per 100000
- Pre-treatment MME trend (slope from regressing MME on year)

Matching Procedure

We used logistic regression to estimate each county's probability of being in Florida based on standardized covariates. We then performed nearest-neighbor matching without replacement: each Florida county was matched to the control county with the closest propensity score.

Results

We tested four specifications (with/without pre-trend, with/without replacement). Across all specifications, Pennsylvania, Illinois, and Oklahoma consistently emerged as the most frequently matched states. These three states were selected as the control group for the difference-in-differences analysis.

⁵ Deborah Dowell, Kun Zhang, Rita K. Noonan, and Jason M. Hockenberry, "Mandatory Provider Review and Pain Clinic Laws Reduce the Amounts of Opioids Prescribed and Overdose Death Rates," *Health Affairs* 35, no. 10 (2016): 1876–1883; and Katherine Wen et al., "State Policies for Prescription Drug Monitoring Programs and Adverse Opioid-Related Hospital Events," *Medical Care* 58, no. 7 (2020): 610–616.